Inference

Until now, we were talking about the mechanics of describing and summarizing distributions in real life datasets.

The reason we do this is that we want the real life data to teach us something about the larger world. (But not always.)

But, real life datasets are often just snapshots of that larger world.

The process of learning about the larger world using datasets is known as **inference**. There are two main flavors:

1. **Statistical inference**: how can we learn from relationships in datasets when we have some uncertainty about the link between the data and the real world?

Inference

Until now, we were talking about the mechanics of describing and summarizing distributions in real life datasets.

The reason we do this is that we want the real life data to teach us something about the larger world. (But not always.)

But, real life datasets are often just snapshots of that larger world.

The process of learning about the larger world using datasets is known as **inference**. There are two main flavors:

2. **Causal inference**: how can we learn whether relationships in datasets specifically teach us about *causal* relationships in the real world?

Where Does Data Come From?

We've already talked a little about the origins of data.

There's a more abstract question we now need to tackle:

↪ What processes in the world create the conditions under which the data we see comes into being?

For example, when a researcher to runs an experiment, what processes make it possible to observe interpretable patterns in the resulting dataset?

Data Generating Processes

Using Gailmard's definition, a **data generating process** (or **DGP**) "is a rule or set of rules governing the social or political events that an analyst wishes to study and the rules by which observations of its results come to be represented in a dataset."

Think back to Prof. Hubert's dataset: all civil rights cases filed in seven courts from 1995 to 2016.

He was not specifically and solely interested in these seven courts, these 20 years, or the 200 judges in my dataset.

He collected this data because he wanted to study the relationship between partisan appointment of judges and civil rights outcomes.

He wanted to learn about the larger data generating process.

Data Generating Processes

When one uses a dataset <u>to evaluate theories</u>, there are two "metaphysical commitments" one makes.

1. The dataset isn't all there is: there's a DGP that generated the specific dataset.

2. Data arises from a DGP in one of two ways:

   2.1 A DGP is **deterministic** if the data it generates involves no random chance.

   2.2 A DGP is **stochastic** if the data it generates involves some degree of random chance.

These are starting presumptions; there's no way to empirically evaluate whether they are "true."

Deterministic DGPs

For a deterministic DGP, a researcher typically tries to figure out the **necessary and sufficient conditions** for some event.

Republican	Pro-Plaintiff
1	0
1	0
1	1
0	1
0	1

$\hookrightarrow$ Having a Republican judge is necessary for a pro-defendant outcome. (But it is not sufficient!)

$\hookrightarrow$ Having a Democratic judge is sufficient for a pro-plaintiff outcome. (But it is not necessary!)

Deterministic DGPs

For a deterministic DGP, a researcher typically tries to figure out the **necessary and sufficient conditions** for some event.

Republican	Pro-Plaintiff
1	0
1	0
1	0
0	1
0	1

↪ Having a Republican judge is necessary and sufficient for a pro-defendant outcome.

↪ And equivalently: having a Democratic judge is necessary and sufficient for a pro-plaintiff outcome. (Why is this equivalent?)

Stochastic DGPs

In last example, why would we assume a deterministic DGP?

$\hookrightarrow$ Do we really think that there's no random chance in the way judges make decisions on cases??

There are three possible sources of uncertainty that would lead someone to assume a stochastic DGP:

1. **Sampling uncertainty**: the data is a **probability sample** of the larger population of interest (controlled by researcher)
2. **Theoretical uncertainty**: the positive theory being examined has missing elements, which could be important for the DGP
3. **Fundamental uncertainty**: exactly what it sounds like...

Stochastic DGPs

A few important notes on this:

- ↪ Sampling uncertainty is controlled by the researcher, so it's pretty easy to deal with in our statistical analysis.
- ↪ Most(?) of the time, we're dealing with theoretical or fundamental uncertainty.
- ↪ Theoretical uncertainty is "good": it's a byproduct of the fact that <u>all</u> theories are simplifications of the world.
- ↪ It's hard to know exactly how (and whether) theoretical and fundamental uncertainty manifest in real life data.

Stochastic DGPs

Core dilemma: you typically only observe one realization of a DGP (i.e., one dataset), but to do statistical analysis, we need to have some idea how the uncertainty works.

This is easy when a researcher controls how the uncertainty arises in her data, such as with survey research.

$\hookrightarrow$ In this case, we say the dataset arises by way of **design-based sampling**, in which sampling uncertainty is the issue.

But, when a researcher does not control how the uncertainty arises in her data, then she has to make assumptions about (i.e., "model") the uncertainty. This is much harder.

$\hookrightarrow$ The dataset arises by way of **model-based sampling**.

Stochastic DGPs

In the remainder of this course, we'll assume stochastic DGPs.

Most researchers are comfortable assuming that real life social processes involve some random chance.

But so far, we've been pretty abstract about what exactly a DGP is.

We'll conceptualize them as **probability distributions**.

$\hookrightarrow$ Review the basics of probability from math camp.

Mathematical Representations of DGPs

Recall: DGPs are the way we talk about the larger social and political processes that generate real life data.

We always like to try to simplify complex realities into models. And we will model DGPs using probability distributions.

A **probability distribution** describes a *hypothetical* (or "theoretical") distribution that arises from a random process.

This is distinct from an **empirical distribution** (or **frequency distribution**) which you've already seen.

We think of empirical distributions as being real-life manifestations of probability distributions (which we cannot directly observe).

Mathematical Representations of DGPs

Quick aside: the following material in topics 4A and 4B will be relatively more mathematical than in topics 1–3

- ↪ This material is more foundational in nature – a lot of it is not directly used in applied research.
- ↪ However, it is important to understand the foundations of probability and statistics to be able to understand the applied research.
- ↪ Still, if some of the mathematical details are confusing, don't worry too much.
- ↪ I will try to highlight the key takeaways from the math.

Probability Spaces/Models

We're all intuitively familiar with the idea of randomness.

But we need to attach some vocabulary, math and notation to our intuitions so we can be more precise.

We'll do this using a simple example: rolling a 6-sided die.

When you roll a die, one of six numbers will be facing up.

$\hookrightarrow$ We refer to the specific roll as an **experiment***.

$\hookrightarrow$ We refer to the <u>possible</u> outcomes as the **sample space**.

$\hookrightarrow$ We usually use S to label a sample space. In this example, the sample space can be written: $S = \{1, 2, 3, 4, 5, 6\}$.

Probability Spaces/Models

We use the term **probability** to describe the way that the randomness operates in a particular situation.

This can get a little mathematically complicated, but for our purposes, we simply need to know the probability with which each element of S can be chosen in an experiment.

For example: a roll of a (fair) die will yield each of the possible outcomes with equal probability, probability $\frac{1}{6}$.

I will loosely refer to a probabilistic situation (like a roll of a dice) as a **probability space (or model)**.

$\hookrightarrow$ The key thing for a probability space/model: we need to know the sample space and the probabilities.

Probability Spaces/Models

Here: we are going to jump right into *probability distributions*, which are the link between basic probability and statistics.

Random Variables

When we do statistics, we will typically want to "translate" a sample space S into numbers.

A **random variable** (or RV) is a function that translates outcomes of an experiment (elements of S) into real numbers.

- $\hookrightarrow$ By convention, we use uppercase letters (e.g., X or Y) to label RVs. So, $X : S \to \mathbb{R}$ is the "mathy" way to write an RV called X for a probability model with sample space S.
- $\hookrightarrow$ RVs will only return as many numbers as elements in S. We label that set of numbers with curly capital letters, like $\mathcal{X}$.
- $\hookrightarrow$ The specific value of an RV in an experiment is its **realization**, and we use lower case letters to label realizations, like $x \in \mathcal{X}$.

Random Variables

A random variable representing the probability space for one roll of a fair six-sided die will only return one of six numbers.

So, if we label that random variable X and the sample space S, then: $S = \{1, 2, 3, 4, 5, 6\}$ and $\mathcal{X} = \{1, 2, 3, 4, 5, 6\}$.

But wait! S is already comprised of numbers—why translate?!

$\hookrightarrow$ A bit philosophical: S isn't *really* comprised of numbers.

This is an example of a **discrete random variable X** in which the set $\mathcal{X}$ is countable and finite (i.e., "discrete").

Random Variables

There are some cases where the sample space S is may not be comprised of numbers.

Example: 5-scale of left-right political ideology.

Here, the sample space is defined as

$$S = \{\text{Strong R}, \text{Weak R}, \text{Moderate}, \text{Weak D}, \text{Strong D}\}$$

We can translate this into a random variable that consists of numbers: $X : S \to \mathcal{X}$ where $\mathcal{X} = \{1, 2, 3, 4, 5\}$.

Random Variables

Consider a different random process: spraying water into an empty container many feet away.

$\hookrightarrow$ Assume you spray 1 liter of water toward a 1 liter container.

There is some randomness since spraying water is imprecise. Some of it will not land in the container.

Measuring the amount you get in the container in milliliters, then we have an experiment from a probability space with $S = [0, 1000]$.

If we call the random variable Y, then $\mathcal{Y} = [0, 1000]$.

This is an example of a **continuous random variable** X in which the set $\mathcal{Y}$ is non-countable and infinite ("a continuum").

Probability Distributions

So far, we haven't talked about the randomness of an RV.

For example, we may want to know the probability that a particular value of the RV will be realized.

$\hookrightarrow$ Using math notation, we usually write the probability that an RV labeled X will take a specific value x as $\Pr[X = x]$ or $\Pr[x]$.

$\hookrightarrow$ Some people use round brackets. Such is life.

For any RV, there are two important functions that tell us about how the randomness works.

These functions describe the *probability distribution* for that RV.

Discrete Probability Distributions

A **probability mass function** (or **PMF**) of a <u>discrete</u> random variable X specifies the probability that $X = x$ for all x.

We usually call this function f, so in math: $f : \mathcal{X} \to [0, 1]$.

$\hookrightarrow$ So, you plug in any element x from $\mathcal{X}$, and the PMF will return a probability for that element: $f(x)$.

But, the function has to return "proper" probabilities, meaning:

1. $0 \leq f(x) \leq 1$ for all $x \in \mathcal{X}$.
2. $\sum_{x \in \mathcal{X}} f(x) = 1$.
3. For any subset of possible outcomes, the probability that any one of those outcomes arises is the sum of the probabilities of each outcome.

Discrete Probability Distributions

For example, consider our example: one roll of a fair die.

Then, we can write the PMF of this random variable as:

$$f(x) = 1/6 \quad \text{for all } x \in \mathcal{X}$$

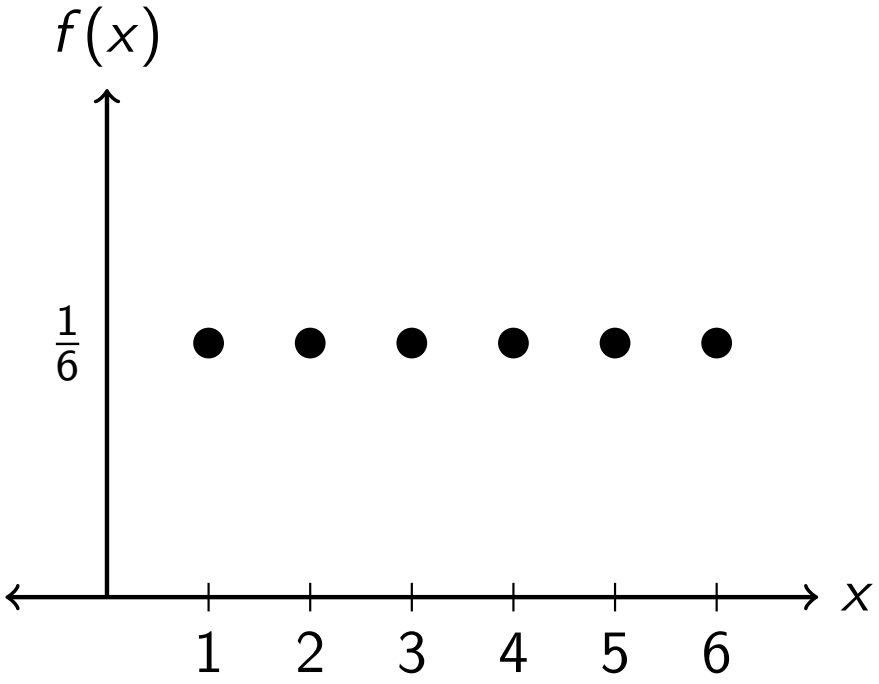
We can verify that this function gives us proper probabilities:

1. For each outcome x, $f(x) = 1/6$ which is between 0 and 1.
2. $f(1) + f(2) + f(3) + f(4) + f(5) + f(6) = \frac{1+1+1+1+1+1}{6} = 1$.

Practical note: property 3 will always be satisfied if properties 1 and 2 are satisfied and X is a random variable.

Discrete Probability Distributions

We can plot this PMF:



Discrete Probability Distributions

A **cumulative distribution function** (or **CDF**) of a discrete random variable specifies the probability that $X \leq x$.

We usually call this function F. So, in math $F : \mathcal{X} \to [0, 1]$ where:

$$F(x) = \Pr[X \leq x] = \sum_{a \leq x} f(a)$$

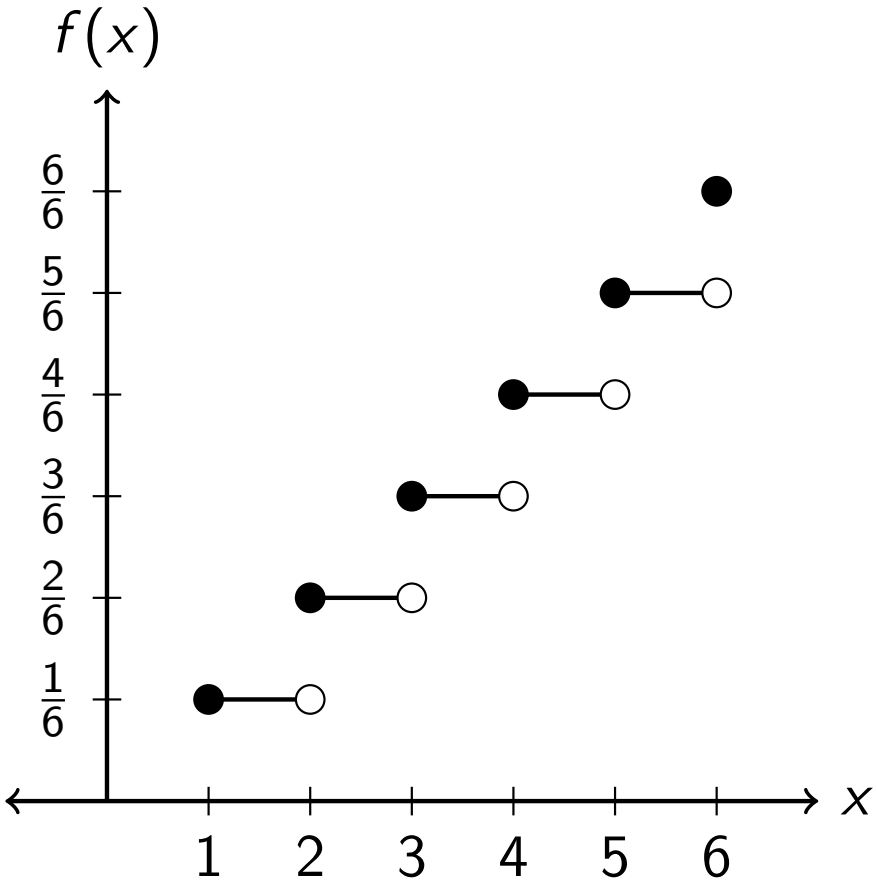
For example, the CDF of our die roll random variable is $F(x) = \frac{x}{6}$.

Suppose we want to know the probability of rolling a 4 or below.

We can calculate it as:
$F(4) = \Pr[X \leq 4] = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{4}{6} = \frac{2}{3}$.

Discrete Probability Distributions

We can plot this CDF:



Summary so far

Random variables: functions that translate outcomes of an experiment (elements of S) into real numbers.

Basically a more "theoretical" way to think about variables that we observe in real life.

$\hookrightarrow$ How do we know how likely each outcome x of RV X is?

$\hookrightarrow$ For discrete RVs, we have probability mass functions (PMFs) and cumulative distribution functions (CDFs).

Summary so far

Random variables: functions that translate outcomes of an experiment (elements of S) into real numbers.

Basically a more "theoretical" way to think about variables that we observe in real life.

- $\hookrightarrow$ How do we know how likely each outcome x of RV X is?
- $\hookrightarrow$ For discrete RVs, we have probability mass functions (PMFs) and cumulative distribution functions (CDFs).
- $\hookrightarrow$ Probability mass functions: a function $f(x)$ that tells us the probability $Pr(X = x)$.
- $\hookrightarrow$ Cumulative distribution functions: a function $F(x)$ that tells us the probability $Pr(X \leq x)$.

Continuous Probability Distributions

Things get mathier when we turn to continuous random variables.

The basic "problem" (although not really) is that the set $\mathcal{X}$ has an infinite number of possible outcomes.

Example: Amount of ice cream in one scoop – can be 1 oz, 1.1 oz, 1.11 oz etc. – infinitely many values.

Two thorny mathematical questions arise from this:

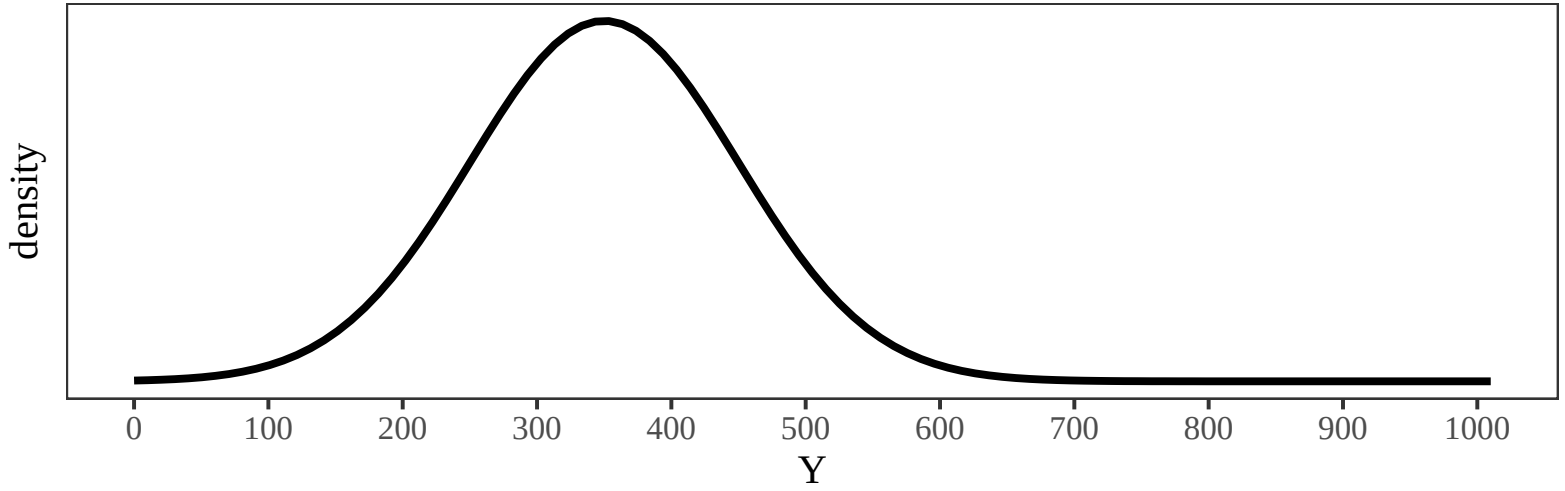
1. For the PMF: how do we come up with "proper" probabilities for each possible $x \in \mathcal{X}$ if there are infinitely many of them?
2. For the CDF: how can we possibly sum together an uncountable set of objects?

These are the kinds of issues that calculus is meant to deal with.

Continuous Probability Distributions

The **probability density function** (or **PDF**) of a continuous random variable X specifies the density for all $x \in \mathcal{X}$.

Roughly: a PDF gives us the "shape" of a probability distribution.



We don't need to worry too much about the technical details of density, except note: $\Pr[X = x] = 0$ for all x in a continuous RV.

Continuous Probability Distributions

The **support** of a random variable is all the $x \in \mathcal{X}$ where $f(x) > 0$.

$\hookrightarrow$ For a discrete RV, the support of that RV is all the values of x that arise with strictly positive probability.

$\hookrightarrow$ For a continuous RV, the support of that RV is all the values of x that have strictly positive density.

Continuous Probability Distributions

The **cumulative distribution function** of a <u>continuous</u> RV also specifies the probability that $X \leq x$ for all x.

Because we have a continuum, we can't sum. We integrate:

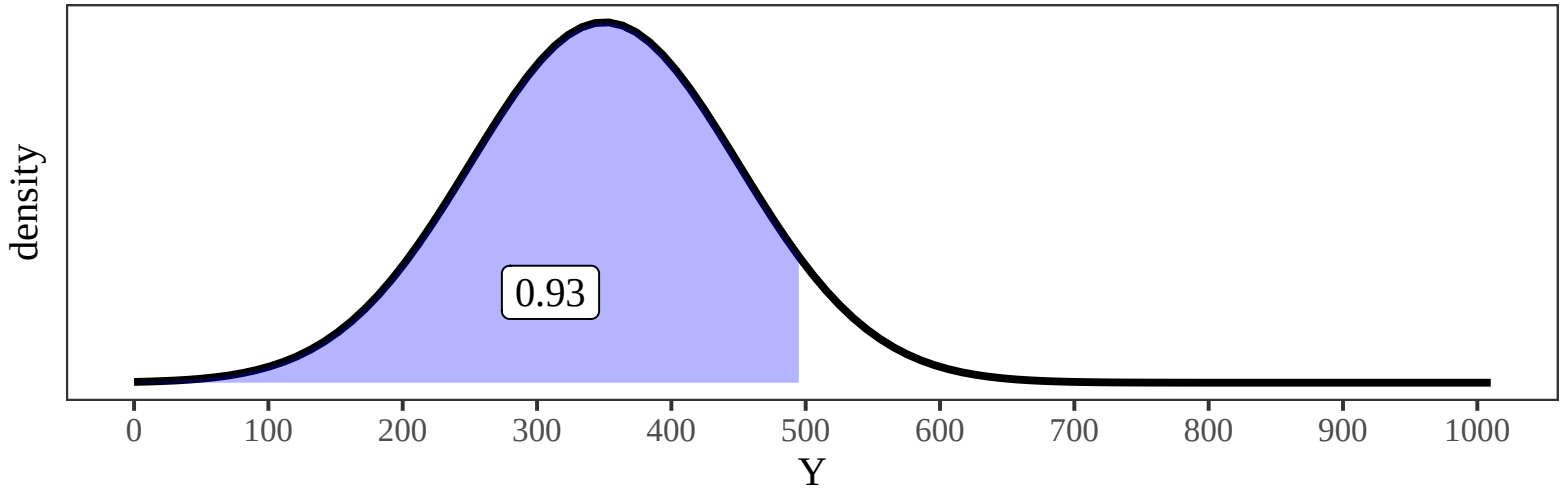
$$F(x) = \Pr[X \leq x] = \int_{a \leq x} f(a)\ da$$

There's a nice geometric way to think about this:

- ↪ Recall from calculus that integrating a function gives the area under that function's curve.
- ↪ Here, we're integrating the PDF, so the CDF gives us the area under the PDF.

Continuous Probability Distributions

For example, what's the probability the container will be less than half full after we spray water toward it?



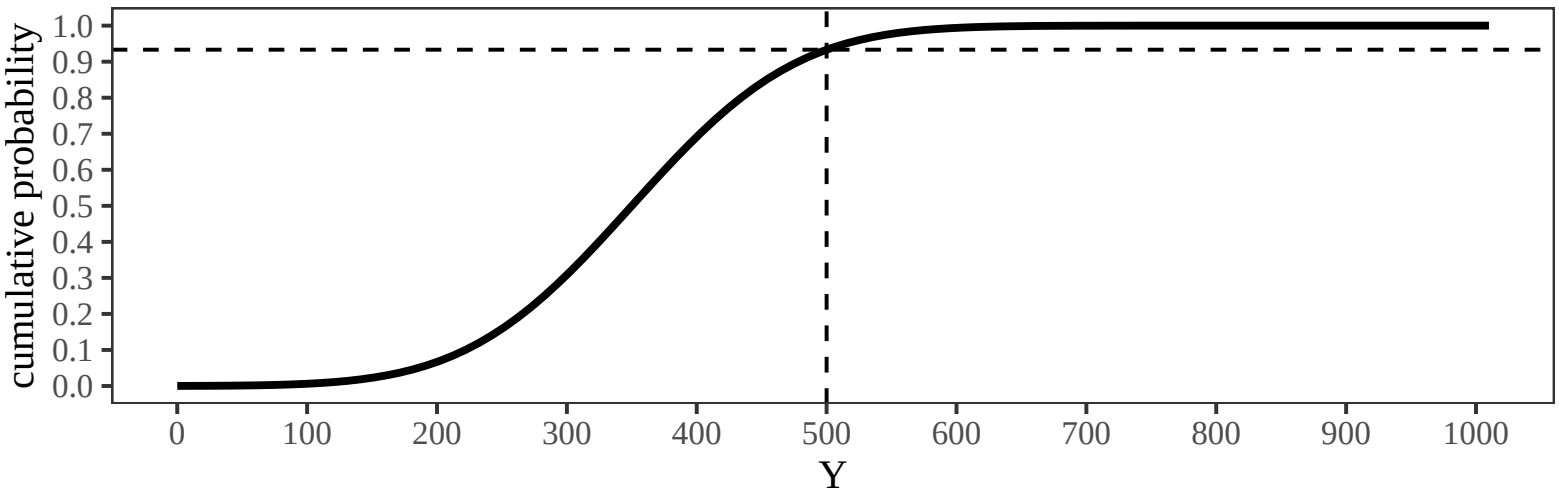
How did I calculate $F(500) = \Pr[Y \leq 500] = 0.93$?

You *could* calculate by hand the definite integral of this PDF from 0 to 500, but I just asked R to do it for me. More on this later.

Continuous Probability Distributions

The last plot was of the PDF, and I was just showing you the area under the curve corresponding to a particular cutoff (500).

We can also directly plot a continuous CDF showing what the area under the curve would be for each possible cutoff:



Continuous Probability Distributions

Some important facts about all CDFs:

- ↪ A CDF ends at one. A CDF for a continuous RV starts at zero. Why?
- ↪ If you want the probability of some number *higher than* x, you can just calculate $\Pr[X > x] = 1 - F(x)$.
- ↪ If you want the probability of some number *between two numbers* x_1 and x_2, you can just calculate

$$\Pr[x_1 \leq X \leq x_2] = F(X \leq x_2) - F(X \leq x_1)$$

When you are calculating things with $<$ or $\leq$, be careful to think about whether your RV is discrete or continuous.

Generally, with continuous RVs, we always work with some form of interval – we never calculate $Pr[X = x_1]$ (which is just 0)

Continuous Probability Distributions

A CDF will be a continuous function if the RV is continuous. (Almost by definition, a discrete RV has a CDF that's not continuous.)

If we have a continuous RV and we assume its CDF is "differentiable everywhere," then:

- ↪ You can get the PDF by differentiating the CDF.
- ↪ You can get the CDF by calculating the definite integral of the PDF from $-\infty$ to x.

Multivariate Probability Distributions

So far, we've talked only about probability distributions for one random variable.

As we discussed in Topic 3, most social scientists are interested in relationships between variables.

Just like we could empirically examine multivariate empirical distributions, we can model their DGPs with multivariate probability distributions.

As we did in Topic 3, we'll build the main ideas for bivariate probability distributions. But, they can extend to distributions with more variables.

Multivariate Probability Distributions

To model the idea that there's a relationship between two variables, we use the idea of joint distributions.

A **joint distribution of X and Y** specifies the probabilistic behavior of each variable and the relationship between them.

- ↪ Again, we typically use f to indicate a PMF/PDF and F to indicate a CDF of a joint distribution.
- ↪ These functions will now have multiple arguments (specifically, two for a bivariate distribution): $f(x, y)$ and $F(x, y)$.
- ↪ They still do the same thing: f takes a value x and a value y and returns a probability/density, and F takes a value x and a value y and returns a cumulative probability.

Multivariate Probability Distributions

A joint PMF/PDF and a joint CDF each return values for combinations of X and Y realizations.

For example, $f(1, 7)$ gives the probability/density of a joint distribution when the X RV is 1 and the Y RV is 7.

We will build ideas with an example of a simple discrete RV (next slide), but all of these concepts apply to continuous RVs too.

Multivariate Probability Distributions

Consider an example in which someone flips a fair coin twice, and calculates these quantities:

$$X = \begin{cases} 0 & \text{if } TT \\ 1 & \text{if } HT \text{ or } TH \\ 2 & \text{if } HH \end{cases} \qquad Y = \begin{cases} 0 & \text{if } HH \text{ or } TT \\ 1 & \text{if } HT \text{ or } TH \end{cases}$$

Do you agree these are discrete random variables?

	$Y = 0$	$Y = 1$
$X = 0$		
$X = 1$		
$X = 2$		

Multivariate Probability Distributions

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Do you agree these are discrete random variables?

	$Y = 0$	$Y = 1$
$X = 0$		0
$X = 1$	0	
$X = 2$		0

Multivariate Probability Distributions

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Do you agree these are discrete random variables?

	$Y = 0$	$Y = 1$
$X = 0$	$\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$	0
$X = 1$	0	
$X = 2$	$\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$	0

Multivariate Probability Distributions

Consider an example in which someone flips a fair coin twice, and calculates these quantities:

$$X = \begin{cases} 0 & \text{if } TT \\ 1 & \text{if } HT \text{ or } TH \\ 2 & \text{if } HH \end{cases} \qquad Y = \begin{cases} 0 & \text{if } HH \text{ or } TT \\ 1 & \text{if } HT \text{ or } TH \end{cases}$$

Do you agree these are discrete random variables?

	$Y = 0$	$Y = 1$
$X = 0$	$\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$	0
$X = 1$	0	$\frac{1}{2} \times \frac{1}{2} + \frac{1}{2} \times \frac{1}{2} = \frac{1}{2}$
$X = 2$	$\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$	0

Multivariate Probability Distributions

Consider an example in which someone flips a fair coin twice, and calculates these quantities:

$$X = \begin{cases} 0 & \text{if } TT \\ 1 & \text{if } HT \text{ or } TH \\ 2 & \text{if } HH \end{cases} \qquad Y = \begin{cases} 0 & \text{if } HH \text{ or } TT \\ 1 & \text{if } HT \text{ or } TH \end{cases}$$

Do you agree these are discrete random variables? In math:

$$f(x, y) = \begin{cases} 1/2 & \text{if } x = 1, y = 1 \\ 1/4 & \text{if } x = y = 0 \text{ or } x = 2, y = 0 \\ 0 & \text{otherwise} \end{cases}$$

The CDF is straight-forward to calculate, but ugly to write.

Multivariate Probability Distributions

The PMF/PDF and the CDF for a joint distribution needs to follow the same rules as for a univariate distribution.

To calculate the CDF for a bivariate joint distribution, you sum/integrate over both variables:

$$\underbrace{F(x,y) = \sum_{a \leq x} \sum_{b \leq y} f(a,b)}_{\text{for discrete RVs}} \quad \underbrace{F(x,y) = \int_{a \leq x} \int_{b \leq y} f(a,b)\ da\ db}_{\text{for continuous RVs}}$$

An example from the previous distribution:

$$\begin{aligned} F(1,1) = \Pr[X \leq 1, Y \leq 1] &= f(0,0) + f(0,1) + f(1,0) + f(1,1) \\ &= \frac{1}{4} + 0 + 0 + \frac{1}{2} = \frac{3}{4} \end{aligned}$$

Multivariate Probability Distributions

Important note: the joint PMF/PDF and the joint CDF tell us about the *relationship* between the two variables.

$\hookrightarrow$ If the two variables are related, it will be apparent from looking at their joint PMF/PDF/CDF.

Compare two extreme examples of joint PMFs over two RVs:

no correlation

	$Y = 0$	$Y = 1$
$X = 0$	$\frac{1}{4}$	$\frac{1}{4}$
$X = 1$	$\frac{1}{4}$	$\frac{1}{4}$

perfect negative correlation

	$Y = 0$	$Y = 1$
$X = 0$	0	$\frac{1}{2}$
$X = 1$	$\frac{1}{2}$	0

Marginal Distributions

A joint distribution tells us about the relationship between random variables, but it also tells us about each variable separately.

The **marginal distribution of X** gives us the distribution of X, ignoring how X varies with Y. Formally:

$$\underbrace{f_X(x) = \sum_{y \in \mathcal{Y}} f(x, y)}_{\text{for discrete RVs}} \qquad \underbrace{f_X(x) = \int_{y \in \mathcal{Y}} f(x, y)\ dy}_{\text{for continuous RVs}}$$

Important: if we start with joint distribution, we can derive each variable's marginal distribution. But we cannot go the other way.

$\hookrightarrow$ Going from marginals to a joint distribution is called **ecological inference**, and it creates a whole host of problems.

Marginal Distributions

Think back to our first example of a joint distribution:

	$Y = 0$	$Y = 1$
$X = 0$	$\frac{1}{4}$	0
$X = 1$	0	$\frac{1}{2}$
$X = 2$	$\frac{1}{4}$	0

$$f_X(x) = \begin{cases} \dfrac{1}{4} & \text{if } x = 0 \text{ or } x = 2 \\ \dfrac{1}{2} & \text{if } x = 1 \end{cases} \qquad f_Y(y) = \frac{1}{2} \text{ for all } y \in \mathcal{Y}$$

Conditional Distribution

Marginal distributions tell us the distribution of a particular variable *ignoring* the other variable(s).

In contrast, the **conditional distribution of X given Y** gives you a distribution over X when holding Y fixed at specific values. Formally:

$$f_{X|Y}(x|y) = \frac{\overbrace{f(x,y)}^{\text{joint}}}{\underbrace{f_Y(y)}_{\text{marginal for } Y}}$$

Conditional Distribution

Recall from basic probability that **Bayes's rule** tells us how prior information and new information can be combined to create an updated assessment.

Applying it to probability distributions, we have:

$$\underbrace{f_{Y|X}(y|x)}_{\text{posterior}} = \frac{\overbrace{f_{X|Y}(x|y)}^{\text{likelihood}}\overbrace{f_Y(y)}^{\text{prior}}}{f_X(x)}$$

Practically speaking, Bayes's rule is very useful for reversing the "direction" of the conditional probability.

Conditional Distribution

Why would someone want to know the conditional distribution?

Because this is where you can most easily see the relationship between the variables!

The conditional distribution tells you how the distribution over one variable changes as the other variable changes.

$\hookrightarrow$ The distribution over the first variable will only change as the second variable changes <u>if</u> they are related!

If they are not related, we say that they are **(stochastically) independent**. Formally, $X \perp\!\!\!\perp Y$ if and only if:

$$f(x, y) = f_X(x) f_Y(y) \implies f_{X|Y}(x|y) = f_X(x) \text{ and } f_{Y|X}(y|x) = f_Y(y)$$

Conditional Distribution

Think back to our first example of a joint distribution:

	$Y = 0$	$Y = 1$
$X = 0$	$\frac{1}{4}$	0
$X = 1$	0	$\frac{1}{2}$
$X = 2$	$\frac{1}{4}$	0

$\hookrightarrow$ Let's consider the case of X conditional on $Y = 0$

$\hookrightarrow$ What is $f_Y(Y = 0)$?

$\hookrightarrow$ $f_Y(Y = 0) = \frac{1}{2}$ (i.e. half of the time, Y is equal to 0

We know that the conditional distribution for $Y = 0$ is defined as:

$$f_{X|Y}(x|Y = 0) = \frac{f(x, Y = 0)}{f_Y(Y = 0)}$$

Conditional Distribution

	$Y = 0$	$Y = 1$
$X = 0$	$\frac{1}{4}$	0
$X = 1$	0	$\frac{1}{2}$
$X = 2$	$\frac{1}{4}$	0

We can just do this for all possible values of X:

$$f_{X|Y}(X = 0|Y = 0) = \frac{f(X = 0, Y = 0)}{f_Y(Y = 0)} = \frac{1/4}{1/2} = \frac{1}{2}$$

And so on - we get:

$$f_{X|Y}(x|Y = 0) = \begin{cases} 1/2 & \text{if } x = 0 \\ 0 & \text{if } x = 1 \\ 1/2 & \text{if } x = 2 \end{cases}$$

Summarizing Probability Distributions

Recall from Topics 2 and 3, we wanted to *summarize* empirical distributions because we wanted to reduce complexity.

We summarized them with *central tendency* and *dispersion*.

We'll do the same thing for "theoretical" probability distributions!

The ideas will be very similar, but will involve some different notation and more abstract mathematical formulas.

To reiterate: we are moving from real world data to theoretical distributions of how data could hypothetically be generated.

$\hookrightarrow$ So, summaries of probability distributions involve *expectations* about what kind of data they would hypothetically generate.

Central Tendency of Probability Distributions

The **expected value** of a <u>discrete</u> random variable X is:

$$E(X) = \sum_{x \in \mathcal{X}} x f(x)$$

The **expected value** of a <u>continuous</u> random variable X is:

$$E(X) = \int_{x \in \mathcal{X}} x f(x) \, dx$$

Note that $E(X)$ in the continuous case looks quite similar – just instead of a sum, we use an integral.

Central Tendency of Probability Distributions

Roughly speaking, you can think about the **expectation operator** $E(\cdot)$ as telling you to take a (weighted) average.

$\hookrightarrow$ Here is a basic example from a discrete PMF:

$$f(x) = \begin{cases} 1/3 & \text{if } x = 0 \\ 1/3 & \text{if } x = 1 \\ 1/3 & \text{if } x = 2 \end{cases}$$

What is the expectation?

Central Tendency of Probability Distributions

↪ Here is a basic example from a discrete PMF:

$$f(x) = \begin{cases} 1/3 & \text{if } x = 0 \\ 1/3 & \text{if } x = 1 \\ 1/3 & \text{if } x = 2 \end{cases}$$

What is the expectation?

$$E(X) = \frac{1}{3} \cdot 0 + \frac{1}{3} \cdot 1 + \frac{1}{3} \cdot 2 = \frac{1}{3} + \frac{2}{3} = 1$$

We simply multiply all possible values of X by $Pr(X = x)$, which are given above.

↪ $E(X)$ is the value of g that minimizes MSE: $E((X - g)^2)$.

Central Tendency of Probability Distributions

Expectations have some nice properties:

1. The expected value of a constant a is that constant: $E(a) = a$.
2. The expected value of a constant b times an RV X is that constant times the expected value of X: $E(bX) = bE(X)$.
3. Given two random variables, X and Y, $E(X + Y) = E(X) + E(Y)$.
4. $E(X)$ always falls between the minimum and maximum values that X can take: $\min \mathcal{X} \leq E(X) \leq \max \mathcal{X}$.

One way to summarize the properties 1 through 3 is that the expectation is a **linear operator**.

Central Tendency of Probability Distributions

Consider two random variables, depicted in this table:

	0	1	2	3
X_1	$\frac{5}{12}$	0	$\frac{1}{4}$	$\frac{1}{3}$
X_2	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$

What are $E(X_1)$, $E(X_2)$ and $E(X_1 + X_2)$?

$\hookrightarrow$ We will not do the actual calculation, but how would we do this if we wanted to?

Central Tendency of Probability Distributions

What if we wanted to transform a random variable by applying a function to it?

$\hookrightarrow$ For example, taking the RV X and doing: $\sqrt{X}$ or X^2.

For any function g of a <u>discrete</u> RV X:

$$E(g(X)) = \sum_{x \in \mathcal{X}} g(x) f(x)$$

And for any function g of a <u>continuous</u> RV X:

$$E(g(X)) = \int_{x \in \mathcal{X}} g(x) f(x) \, dx$$

Dispersion of Probability Distributions

The **variance** of a random variable X is:

$$Var(X) = E((X - E(X))^2)$$

Some people label variance as: $V(\cdot)$, $\mathbb{V}(\cdot)$, without italics $\mathrm{Var}(\cdot)$, with square brackets or as σ^2 (Greek "sigma").

Using some algebra and the properties of expectations, we can write the variance in a different way:

$$\begin{aligned} Var(X) &= E((X - E(X))^2) \\ &= E(X^2 - 2XE(X) + E(X)^2) \\ &= E(X^2) - 2E(X)E(X) + E(X)^2 \\ &= E(X^2) - E(X)^2 \end{aligned}$$

Dispersion of Probability Distributions

The variance of a random variable also has nice properties:

1. The variance of a constant a is zero: $Var(a) = 0$.
2. $Var(X)$ is always weakly positive: $Var(x) \geq 0$.
3. The variance of a constant b times an RV X is that constant squared times the variance of X: $Var(bX) = b^2 Var(X)$.
4. If X and Y are stochastically independent, then $Var(X + Y) = Var(X) + Var(Y)$.

Variance is <u>not</u> a linear operator.

Other Measures of Central Tendency

You can also calculate the mean and mode of a probability distribution.

The median of a random variable X is just a value x_M such that:

$$F(x_M) = \frac{1}{2}$$

The mode of a random variable X is just a value x_D where:

$$f(x_D) = \max_{x \in \mathcal{X}} f(x)$$

Does the last expression (for the mode) make sense for continuous random variables?

Practically speaking, we tend to talk less about medians and modes in the context of theoretical probability distributions.